End-to-End Computer Vision Data Pipeline with CVAT Annotation platform

CS 4273 Capstone Design Project – Spring 2026

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OBJECTIVE

1. An end-to-end platform to ingest, process, and manage driving data for compute vision workflows for Open Pilot.
2. A unified system with job execution, artifact storage, and Computer Vision Annotation Tool (CVAT) labeling for future machine learning and human review.

INTRODUCTION

- Computer vision systems need large-scale and structured datasets for training and evaluation
- Raw driving data and footage must be organized into units for processing and analysis
- A batch processing pipeline oriented to both automated processing and human review is needed for efficient computation operations
- Complex structure and services condensed into an intuitive UI design
- The end products address these challenges by providing a platform for data ingestion, processing, storage, and annotation.
- Considered bias in dataset quality and fairness
- Protect sensitive data access and prevent misuse with data management practices.

METHODOLOGY

- An interface for interacting with services: download, control ingestion, processing, jobs, and CVAT annotation
- Extract frame and metadata to store to PostgreSQL and MinIO as an S3 bucket
- Repositories layer to provide abstraction to the platform.
- OpenPilot Download Worker downloads videos and logs from Open Pilot
- Alembic Worker used for data migrations
- OpenPilot Upload Worker reads extracted segments from a shared volume, uploads frames/logs to MinIO, and cleans up local storage
- CVAT workers bridge the database and CVAT server, managing task creation and syncing job statuses
- Pulls from the database and proceeds with automated annotation or human annotations

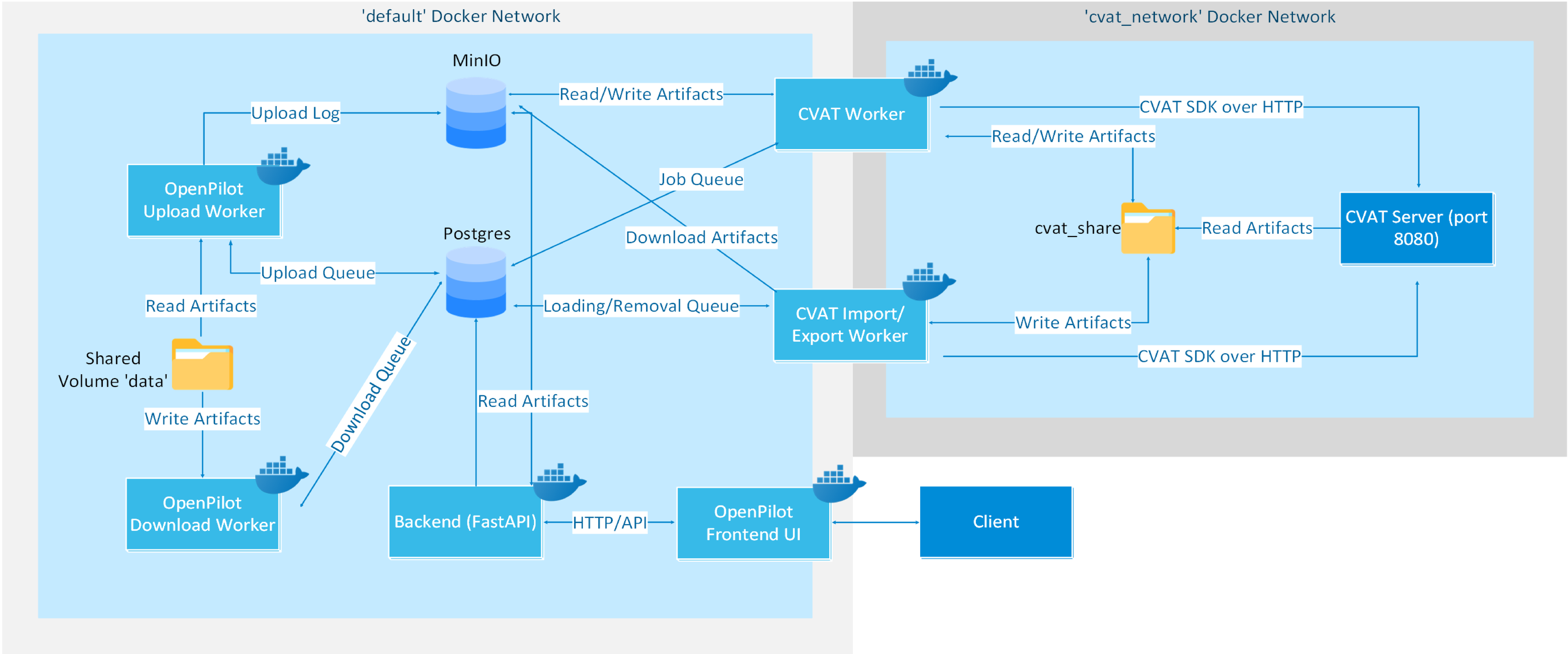

Figure 1: Diagram of the current system architecture

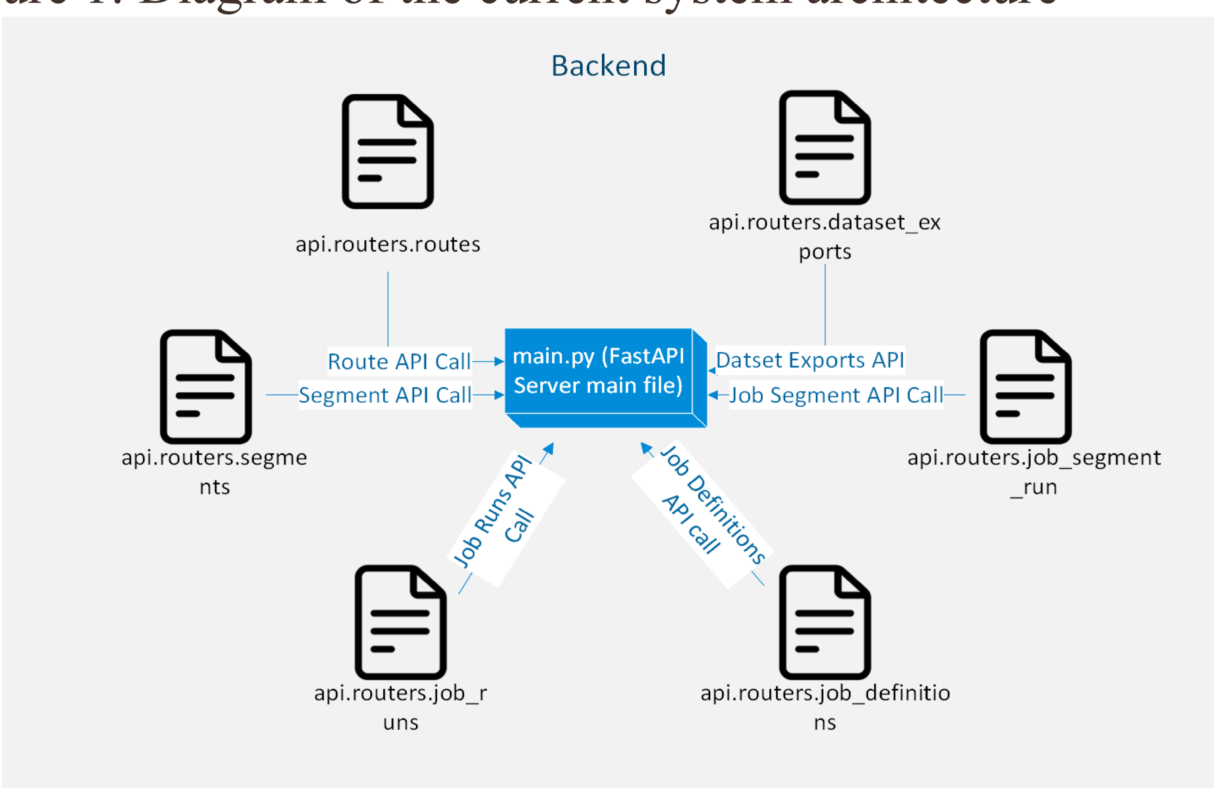

Figure 2: Diagram of backend detail architecture

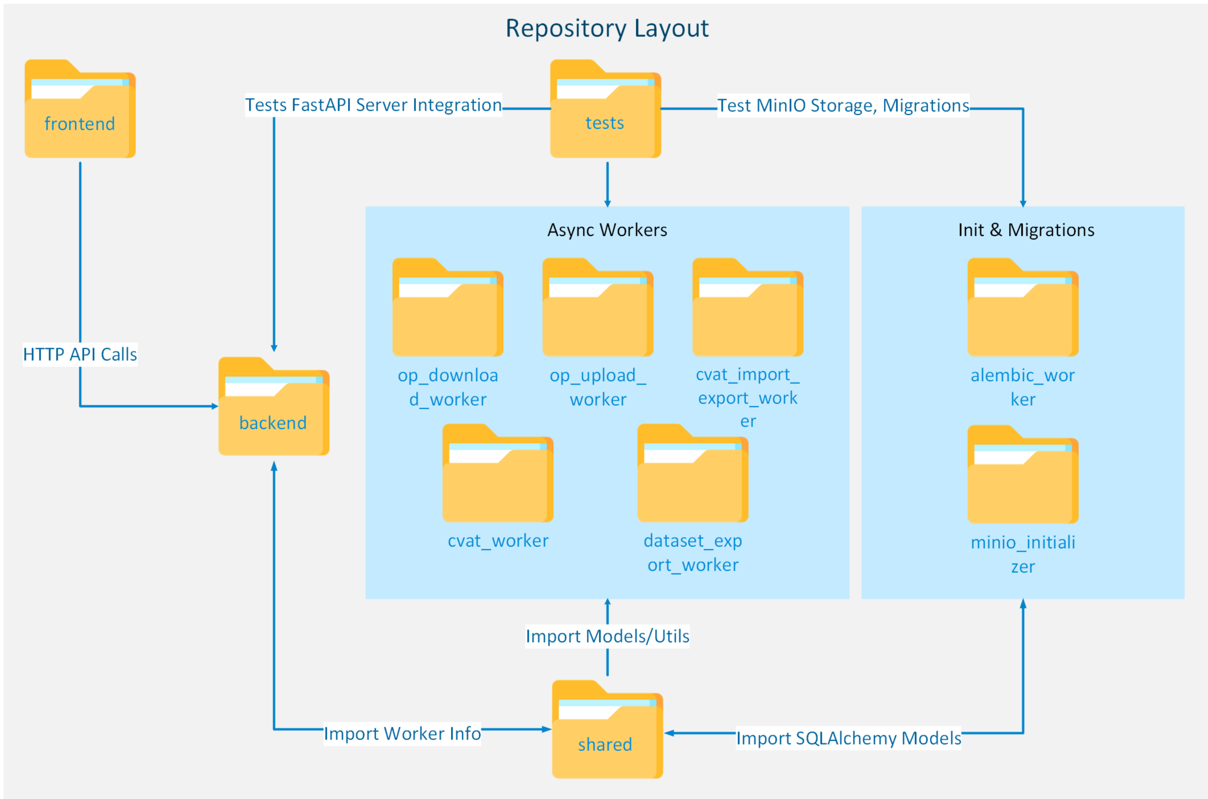

Figure 3: Diagram of repository system architecture

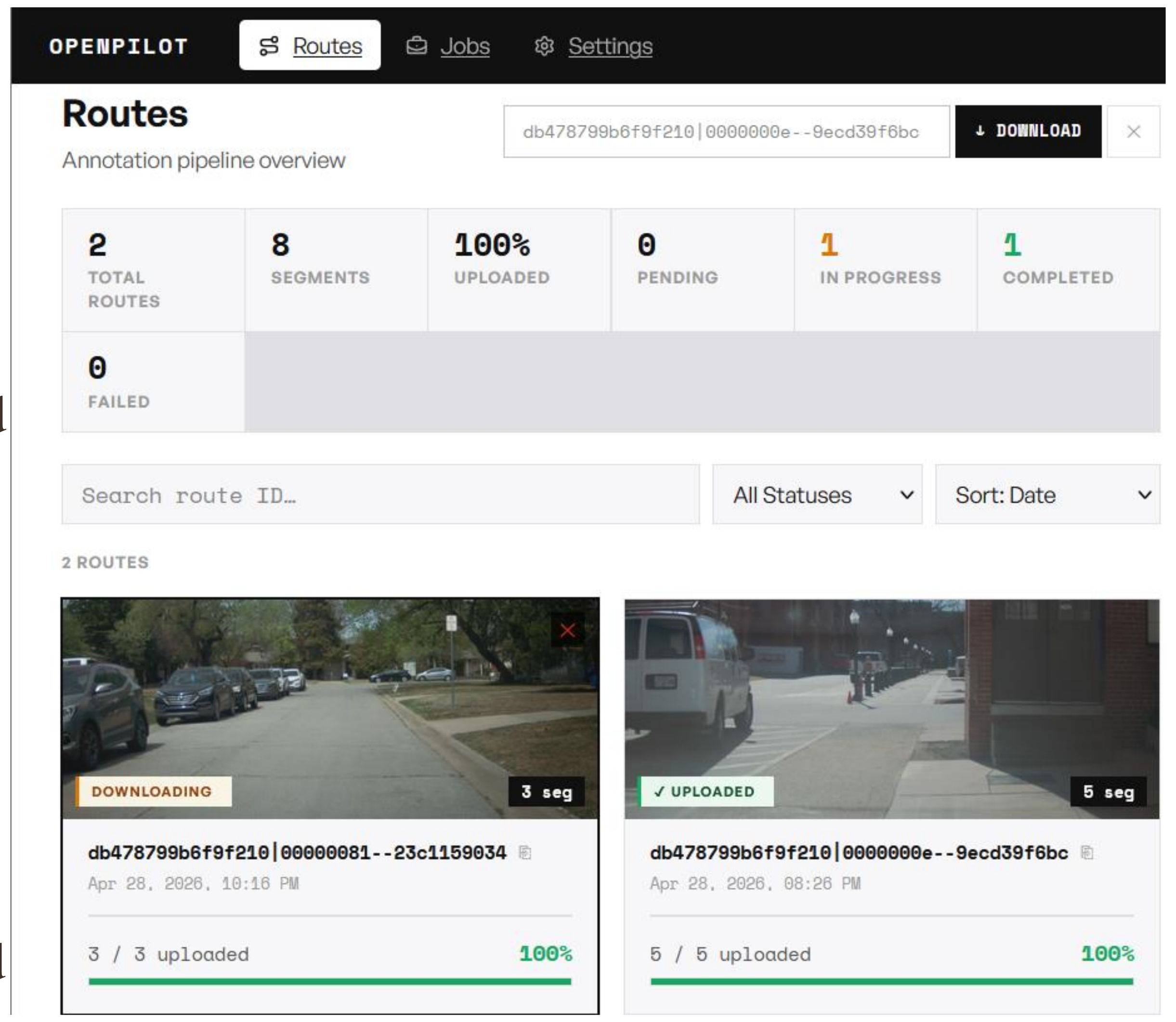

Figure 4: Home page dashboard show all routes

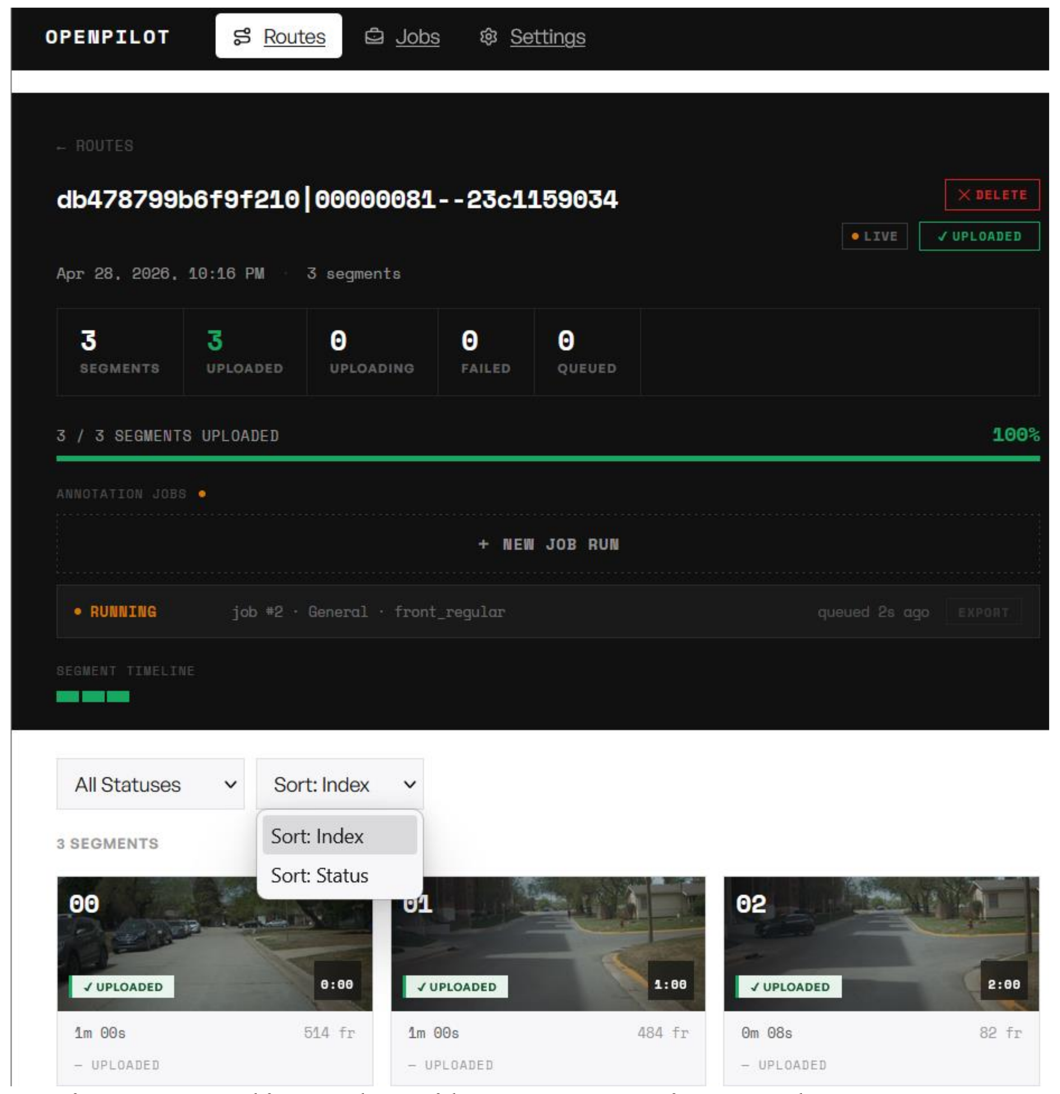

Figure 5: live detail segment in each route page

TECHNOLOGIES USED:

TS comma.ai PostgreSQL FastAPI

RESULTS

- Fully dockerized for deployability capabilities, from the source to a wide range of machines.
- Responsive UI site allows users to perform seamlessly with live updates.
- Batch processing pipeline allows organizing CVAT annotation for automated or human review.
- Schema designs and CRUD operations allow optimized search operations without performance degradation.
- Handles privacy concerns of sensitive driving data responsibly.
- Ensures handling of the comma AI user token safely when taking driving data and artifacts.

CONCLUSION

- The platform streamlines the transition of raw video ingestion input to annotated artifacts.
- Overcame the challenges of having a large amount of video data by implementing a robust backend with distributed workers and a scalable database using MinIO and Postgres.
- The platform provides layout services and storage solutions for later machine learning development.
- Provides a future development path between machine learning processing and human annotation via CVAT integration.
- Recognized biased datasets and need balanced data representation.
- Address data ownership of Comma AI for the generated parameter.

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